STEAMCITY CURRICULUM

NEGAWATT SCENARIO AND ENERGY SOBRIETY

Understanding Energy Through History

theme: energy and energy efficiency

Co-funded by the European Union

Introduction

The history of our energy consumption reveals an acceleration since the industrial revolution. While a European consumed around 15-25 GJ of energy per year at the beginning of the 19th century, this consumption has increased six to tenfold depending on the country, reaching 100-200 GJ in 2020 (Eurostat, 2023). This increase is explained by the mechanization of tasks, widespread electrification, the development of motorized transport, and the rise of domestic appliances.

The négaWatt scenario, developed by the négaWatt Association since 2003, offers a methodical approach to reconciling well-being and drastically reducing energy consumption. It is structured around three complementary pillars: energy sobriety (reducing needs), energy efficiency (optimizing uses), and renewable energies (decarbonizing supply). The sobriety component, at the heart of this protocol, does not mean deprivation but rather questioning our real needs and optimizing our uses.

This approach is crucial in the face of contemporary climate challenges. The IPCC (2022) highlights that demand-side behavioral changes can reduce emissions by 5 to 20% by 2030. In Europe, the REPowerEU plan (2022) and national transition scenarios demonstrate that a significant reduction in energy consumption by 2050 is possible while maintaining quality of life, notably by reducing inequalities in access to energy services between European countries and regions.

The objective of this protocol is to help students understand that energy efficiency does not imply a return to the Stone Age, but rather a reflection on the intelligent use of energy. By exploring the historical evolution of our consumption and by concretely analyzing the levers of energy efficiency in two key areas (transport and housing), students will develop a nuanced vision of the possibilities for individual and collective action to build an energy-efficient and socially desirable future, in line with European carbon neutrality objectives.

Disciplines

physics
history-geography
mathematics
technology
civic education

Sustainable Development Goals

7 AFFORDABLE AND CLEAN ENERGY

11 SUSTAINABLE CITIES AND COMMUNITIES

12 RESPONSIBLE CONSUMPTION AND PRODUCTION

13 CLIMATE ACTION

Overview

Protocol structure

This activity aims to demonstrate that more energy-efficient lifestyles are not only possible but also bring innovations and improvements to quality of life. The educational approach is based on the observation of our daily environment, comparative historical analysis, and forward-looking thinking.

The activity is structured around four progressive phases:

1. **Step 1: Energy inventory of our environment** - Students identify energy uses in their immediate environment (classroom, school, home) and reconstruct the historical evolution of these uses to understand how our needs have changed.
2. **Step 2: Comparison of yesterday's and today's consumption** - Through the analysis of numerical data and concrete examples, students measure the evolution of energy consumption for identical needs and identify the factors of increase.
3. **Step 3: Exploring the levers of sobriety** - Students explore the possibilities of reducing consumption in the areas of transport and housing, distinguishing between technical solutions and behavioral changes.
4. **Step 4: Reflection on essential uses and positive innovations** - Students personally evaluate their own energy uses and identify the energy innovations that seem most remarkable and beneficial to them.

This progression allows students to develop a nuanced understanding of energy sobriety, starting from concrete observation to lead to personal reflection on their own lifestyles and the prospects for desirable development of our society.

Getting started

Duration: min. 2 sessions (adaptable to 3-4 sessions for more in-depth study)

Difficulty level: *Intermediate - Adaptable from lower secondary school to high school by adjusting the complexity of the analyses and the depth of the reflections* ★★★☆☆

Material needed:

- Step 1: Observation sheets, cameras/smartphones to document energy usage, access to different areas of the establishment
- Step 2: Printed energy consumption data tables, calculators, graphs and historical infographics
- Step 3: Activity sheets on transport and housing, data tables on the energy mix, calculators or simple spreadsheets, materials for creating posters (felt pens, poster paper), access to documentary resources
- Step 4: Individual reflection sheets, material for creative presentation of innovations

Glossary

Keywords and concepts	Definitions
Energy sobriety	An approach that involves reducing energy consumption through changes in behavior, usage and organization (individual and collective) while meeting essential needs.
Energy efficiency	Improving the energy efficiency of equipment and systems to obtain the same service with less energy.
NegaWatt scenario	Energy transition proposal developed by the négaWatt Association, based on the sobriety-efficiency-renewables trilogy.
Energy needs	Service that one wishes to obtain (heating, travel, lighting) distinct from the energy consumption necessary to obtain it.
Energy use	Concrete use of energy for a specific need (heating a room, powering a device, propelling a vehicle).
Final energy consumption	Quantity of energy actually consumed by the end user, after transformation and transport.
Energy intensity	Relationship between energy consumption and economic or social activity or service provided.
Energy resilience	Ability of a system to maintain its essential functions despite disruptions in energy supply.
Energy insecurity	A situation where a household has difficulty meeting its basic energy needs due to insufficient resources or energy-intensive housing.
Energy mix	Distribution of different energy sources in the production of electricity in a territory (nuclear, renewable, fossil).
Load factor	Ratio between the actual production of an installation and its maximum theoretical production (expressed in %).
Optimization under constraints	Mathematical method for finding the best solution while respecting several limits simultaneously.
Controllable sources	Energy sources whose production can be adjusted to demand (nuclear, gas, hydraulic).
Intermittent sources	Energy sources whose production depends on variable natural conditions (wind, solar).
Energy sobriety	An approach that involves reducing energy consumption through changes in behavior, usage and organization (individual and collective) while meeting essential needs.

Protocol

Step 1 - Energy inventory of our environment

Context and description of the problem to be solved at this stage: Our daily lives are imbued with energy uses that are often invisible because they seem so natural to us. This first phase invites students to take an analytical look at their immediate environment to become aware of the omnipresence of energy in our activities. By starting from the concrete observation of the spaces they frequent daily, students develop their ability to identify and question energy uses, an essential first step in understanding the challenges of sobriety.

Learning Objectives: Identify energy uses in the everyday environment. Reconstruct the historical evolution of energy needs and uses. Develop a critical perspective on the "normality" of our current consumption patterns.

Conceptualisation

Research question: What are the different energy uses observed in our daily lives and how can we classify them?

- Hypothesis: There are more energy uses in our environment than we immediately perceive, with some being very visible (lighting, heating) and others more discreet (standby of appliances, ventilation).
- Key concepts:
  - Energy use: Specific use of energy to meet a need (definition in the context of direct observation)
  - Energy function: General category of use (allows classification and analysis)
  - Energy requirement: Final service sought (helps distinguish the essential from the superfluous)

Research question: What technical and organizational solutions have been used throughout history to meet basic needs?

- Hypothesis: Pre-industrial societies used solutions that were less energy-intensive but required more time and human effort.
- Key concepts:
  - Energy substitution: Replacement of one form of energy by another (allows understanding of transitions)
  - Energy efficiency: Ratio between service provided and energy consumed (measures technical progress)
  - Structural sobriety: Social organization minimizing energy needs (sheds light on historical choices)

Research question: What technical, economic and social factors have led to the increase and diversification of our energy uses?

- Hypothesis: Access to abundant and inexpensive energy has transformed our lifestyles by creating new needs that are now considered normal.
- Key concepts:
  - Rebound effect: Increase in consumption following efficiency gains (explains continued growth)
  - Energy Dependence: Integration of Energy into All Aspects of Life (Analyzes Societal Transformations)
  - Energy normality: Consumption level considered normal (questions our current standards)

Students Investigation

1. Exploring our energy environment

Objective: To develop an analytical view of the energy uses of our daily environment.

The students, organized into groups, explore different areas of the school (classroom, hallways, cafeteria, courtyard, administrative offices) with the mission of identifying all the energy uses they can observe. They use a simple observation grid:

Place explored:		**Band :**	

Observed energy use	Energy type	Function/need met	Frequency of use
LED ceiling lighting	Electric	See, work	Permanent during the day
Electric radiator	Electric	Heating the space	Depending on the season
Laptop	Electric	Inform, communicate	Several hours/day
...	...	...	...

Each group explores a different space and then meets to share their observations. The teacher helps categorize the identified uses according to four main functions:

- Thermal comfort: heating, air conditioning, ventilation
- Lighting and vision: luminaires, screens, projectors
- Information and communication: computers, telephones, sound systems
- Movement and transport: elevators, vehicles, automatic gates

This first exploration allows students to become aware of the energy density of their daily environment.

2. Time travel: finding the "ancestors"

Objective: To understand the historical evolution of the means of satisfying our basic needs.

For each identified energy use, students collectively research how the same need was met before the advent of current technologies. This research can be based on their own knowledge, materials provided by the teacher, or short guided research sessions.

Examples of historical correspondences:

Current usage	Historical "ancestor"	Observed evolution
Electric lighting	Candles, oil lamps, gas	Light intensity × 1000, 24/7 availability
Central heating	Fireplace, wood stove	Uniform comfort, automatic piloting
Computer	Library, mail, manual calculation	Processing speed × 10^6, near-infinite storage
Elevator	Stairs, ramps	Accessibility, time saving
Fridge	Cellar, pantry, salting room	Long-term preservation, food safety

This historical perspective helps students understand that our basic needs (heating, lighting, communication, transportation) remain constant, but that the means to satisfy them have been transformed by access to abundant and cheap energy.

3. Reflection on transformations

Students collectively reflect on the following questions:

- **What are the benefits of these developments? (comfort, security, speed, accessibility)**
- **What new needs are created? (network dependency, obsolescence, complexity)**
- **Are all these uses essential? (essential/comfort/luxury differentiation)**

This reflection prepares the next phase by questioning the "normality" of our current consumption.

Conclusion & Further Reflexion

The energy inventory allowed students to discover the omnipresence of energy in their daily environment, often invisibly. They understood that our basic needs (heating, lighting, communication, transportation) have remained constant throughout history, but that the means of satisfying them have been revolutionized by access to abundant energy. This transformation has created new comforts but also new energy dependencies.

Direct exploration of familiar spaces encourages student engagement in observation. Comparing historical solutions helps put our current practices into perspective. Classifying them by function structures reflection and facilitates critical analysis of energy uses.

To deepen the reflection and develop a more nuanced understanding of energy issues, here are some key questions to explore collectively:

- **How can we distinguish the essential from the superfluous in our energy uses?** Analyzing vital functions versus additional comforts. Reflecting on the thresholds for satisfying needs. Questioning the new "needs" created by technology.
- What **are the gains and losses of our energy transformations?** Comparison of the advantages (comfort, security, speed) and disadvantages (dependence, complexity, vulnerability) compared to historical solutions. Analysis of the effects on individual and collective autonomy.
- **How are changes in our energy practices transforming our relationship with time and space?** The impact of energy on social organization and lifestyles. Changes in intergenerational relationships and traditional knowledge.
- **What energy alternatives could we consider for the future?** Thinking ahead about the possible evolution of our uses. Imagining scenarios for energy sobriety or efficiency. Preparing to quantify consumption to inform future choices.

Step 2 - Comparison of yesterday's and today's consumption

Context and description of the problem to be solved at this stage: After identifying the qualitative evolution of our energy uses, this phase quantifies these evolutions to concretely measure the extent of the transformations. By manipulating numerical data on energy consumption, students develop their understanding of orders of magnitude and become aware of the recent acceleration of our energy needs.

Learning Objectives: Quantify the evolution of energy consumption. Understand the factors that explain this evolution. Develop the ability to critically analyze energy data.

Conceptualisation

Research question: How has technological evolution changed the energy efficiency of essential services over time?

- Hypothesis: Technological evolution has created divergent trajectories: some services have become more energy efficient, while others have seen their consumption explode despite efficiency gains, due to the increase in usage volumes and the creation of new needs.
- Key concepts:
  - Historical energy efficiency: Evolution of the ratio between service provided and energy consumed over the long term (allows real technical progress to be measured)
  - Energy intensity of a service: Quantity of energy required to obtain a unit of service (lighting, transport, heating) at a given time
  - Technological substitution: Replacing one technology with another for the same service (reveals technical choices and their consequences)

Research question: Why did some services see their energy consumption decrease while others exploded?

- Hypothesis: Services that already existed before the industrial era (lighting, heating, transport) have benefited from technical optimizations, while new services (refrigeration, electronics) have created new consumption items that have become widespread.
- Key concepts:
  - Pre-existing vs. emerging services: Distinction between historical needs and needs created by modernity (explains differentiated trajectories)
  - Democratization effect: Generalization of an initially elite service (multiplies the overall energy impact)
  - Creation of needs: Emergence of new services considered essential (reveals the evolution of living standards)

Research question: What is the relationship between technical progress, comfort of life and total energy consumption?

- Hypothesis: Technical progress allows for local efficiency gains, but the increase in comfort and the multiplication of uses generate an overall growth in energy consumption, creating a paradox between optimization and sobriety.
- Key concepts:
  - Efficiency paradox: Situation where efficiency gains are offset by increased usage (questions the role of technology in sobriety)
  - Comfort standards: Level of energy service considered normal at a given time (reveals the evolution of social expectations)

- Historical rebound effect: Increase in total consumption despite improved efficiency (highlights the limits of all-technology)

Students Investigation

1. Comparative analysis of consumption

Objective: To concretely measure the evolution of energy consumption for identical needs.

Students receive a comparative table of energy consumption for different needs at different times. They work in pairs to analyze this data:

Need	1900	1950	2000	2020	Evolution factor
Lighting a room (1 hour)	Bougie : 0,01 kWh	Ampoule incandescente : 0,1 kWh	Halogen bulb: 0.05 kWh	LED : 0.008 kWh	÷ 1,25
Heating a home (year)	Wood: 15,000 kWh	Charbon: 12,000 kWh	Fuel oil: 20,000 kWh	Gas + isolation: 8,000 kWh	÷ 1,9
Paris-Lyon transport	Steam train: 250 kWh	Electric train: 80 kWh	Car: 45 kWh	TGV: 25 kWh	÷ 10
Washing clothes (5kg)	Manual washing: 2 kWh (human effort)	Washing machine: 3 kWh	Modern washing machines: 1 kWh	Washing machine A+++ : 0.6 kWh	÷ 3,3
Food preservation (year)	Natural cellar: 0 kWh	Cooler: 100 kWh	Refrigerator: 800 kWh	A+++ refrigerator: 150 kWh	× 150

Students calculate the evolution factors and identify trends:

- Improved efficiency: For certain uses (lighting, transport, heating), technology allows less energy to be consumed for the same service
- New energy needs: Certain services (refrigeration, electronics) create new consumption items
- Change of scale: The widespread use of previously rare equipment multiplies total consumption

2. Debate: progress or overconsumption?

Objective: To develop a critical view of the evolution of our energy consumption.

Based on the analyzed data, the teacher organizes a structured debate around the question: "Does the evolution of our energy uses represent progress or a drift towards overconsumption?"

Progress arguments

- Improved comfort and quality of life
- Save time and increase efficiency
- Democratized access to services once reserved for the elite
- Improving safety and health

Overconsumption arguments

- Multiplication of equipment and uses
- Creation of artificial needs through advertising
- Planned obsolescence and accelerated renewal
- Environmental and social impact of energy production

The teacher helps students move beyond the binary opposition by introducing the notion of chosen sobriety: it is possible to benefit from the advantages of technical progress while questioning the necessity of certain uses.

To help students develop their intuition about energy consumption, the teacher can present some simple benchmarks:

- 1 kWh = energy to run: a refrigerator for 2 days, a laptop for 10 hours, or travel 5 km in an electric car
- Average consumption of a French person: 50 kWh per day (all uses combined)
- 1 liter of gasoline = 10 kWh: energy equivalent of a day of human physical work

These benchmarks will allow students to better assess the issues in the following phases.

Conclusion & Further Reflexion

The comparative analysis of energy consumption revealed to the students the complexity of technological evolution trajectories. They discovered that energy efficiency does not follow a linear progression: some services (lighting, transportation) have benefited from remarkable optimizations, while others (refrigeration, electronics) have created entirely new energy needs. This diversity of evolutions explains why our overall consumption continues to increase despite technical progress, illustrating the paradox between local efficiency and global sobriety.

Handling concrete numerical data facilitates understanding of energy orders of magnitude and develops critical thinking in the face of discourse on technical progress. Structured debate allows students to move beyond binary positions to construct a nuanced vision of energy developments. The introduction of quantitative benchmarks (kWh, equivalences) provides students with assessment tools that they can use in subsequent phases.

Students have developed their ability to critically analyze energy data and gained an understanding of the historical mechanisms that have shaped our current consumption. They now have a grasp of the essential orders of magnitude and are able to identify the explanatory factors behind energy developments (democratization, technological substitution, creation of needs).

To deepen the reflection and develop a more systemic understanding of energy issues, here are some key questions to explore collectively:

- **Why isn't energy efficiency enough to reduce our overall consumption?** An analysis of the rebound effect and compensation mechanisms. Understanding the limits of all-technology. Considering the conditions for truly low-carbon energy efficiency.
- **How do comfort standards evolve, and who determines them?** Exploring the social processes of standardizing energy use. Analyzing the role of advertising, public policy, and innovation in creating new needs. Questioning our collective ability to redefine these standards.
- **What lessons can be learned from past energy trajectories to anticipate the future?** Identifying the factors that have promoted or hindered certain historical energy transitions. Applying these lessons to the current challenges of the energy transition. Reflecting on levers for action to guide future developments.

- **How can we reconcile technical progress and energy moderation in our individual and collective choices?** Exploring the conditions for innovation in the service of sobriety. Reflecting on the criteria for evaluating new technologies. Preparing for a concrete analysis of the levers of sobriety in the fields of transport and housing.

Step 3 - Exploring the levers of sobriety

Context and description of the problem to be solved at this stage: With their understanding of the historical and quantitative evolution of our energy uses, students are ready to concretely explore the possibilities of sobriety. This phase focuses on representative areas - transport, housing, food, which together represent approximately 60-70% of the final energy consumption of European households depending on the country. The objective is to show that sobriety does not mean deprivation but intelligent optimization of our uses.

Learning Objectives: Identify the levers of sobriety in transportation and housing. Distinguish between technical solutions and behavioral changes. Evaluate the feasibility and acceptability of different sobriety measures.

Conceptualisation

Research question: What action levers are available to reduce energy consumption without degrading quality of life?

- Hypothesis: Energy sobriety can be achieved through a combination of behavioral changes, technical optimizations and collective reorganizations, with savings potential varying depending on the areas and territorial contexts.
- Key concepts:
  - Energy sobriety: Reduction of consumption by modifying uses without degrading the service provided (distinguished from technical efficiency)
  - Behavioral vs. Structural Leverage: Individual Actions Modifiable in the Short Term vs. Organizational Changes Requiring Collective Transformations
  - Acceptability threshold: Limit beyond which sobriety measures are perceived as a deprivation (determines social feasibility)

Research question: How do territorial and socio-economic constraints influence the possibilities of energy sobriety?

- Hypothesis: The room for maneuver for energy sobriety varies greatly depending on the geographical context (urban/rural), available infrastructure (public transport, networks) and the economic resources of households, creating inequalities in the face of the transition.
- Key concepts:
  - Territorial constraint: Limitation of energy choices imposed by geography and infrastructure (explains the differences in available options)
  - Inequalities in sobriety: Differentiated capacities to adopt sober practices according to resources and contexts (reveals social justice issues)
  - Threshold of imposed vs. chosen sobriety: Distinction between economic constraints and deliberate choices of reduction (questions the voluntary aspect of sobriety)

Research question: What is the relationship between technical solutions (efficiency) and behavioral solutions (sobriety) in reducing consumption?

- Hypothesis: Technical efficiency and behavioral sobriety are complementary but can also be a source of tension: efficiency gains can disinhibit sober behavior (rebound effect), while overly restrictive sobriety can hinder the adoption of technical solutions.
- Key concepts:

- Complementarity efficiency-sobriety: Synergy between technical improvement and modification of uses (maximizes energy savings)
- Behavioral rebound effect: Increase in usage following a reduction in costs or technical improvement (limits the impact of efficiency gains)
- Systemic optimization: Search for the optimal balance between technical efficiency, behavioral sobriety and quality of life (guides integrated strategies)

Students Investigation

1. Transport workshop: rethinking our travel

Objective: Explore the possibilities of reducing energy consumption linked to travel.

The students collectively analyze the "transport budget" of a typical European family, then explore different levers of sobriety.

Reference situation - Typical European family (2 adults, 2 children, peri-urban area)

- 1-2 cars depending on the country (EU average: 1.4 cars/household)
- 15,000-30,000 km/year depending on the geography of the country
- Consumption: 5-8L/100km on average
- 1,000-2,000 L of fuel/year = 10,000-20,000 kWh/year

In small groups, students reflect on the feasibility of these different strategies in different contexts (dense urban vs. rural, developed vs. limited public transport) and imagine the daily lives of typical families in different scenarios. They then present their findings in the form of a "typical day" illustrating the necessary adaptations according to territorial specificities.

Levers of sobriety to explore

Strategy	Description	Potential savings	Conditions for success
Route optimization	Grouping of activities, planning	-20% mileage	Family organization
Report modal	Bicycle/public transport for certain journeys	-30% car use	Adapted infrastructure
Telework	2 days/week for a parent	-40% off commuting	Employer agreement
Car sharing	One car + occasional car sharing	-50% fixed costs	Local offer available
Chosen proximity	Moving closer to work	-70% daily journeys	Real estate opportunities

2. Habitat workshop: optimizing our energy comfort

Objective: To identify opportunities for reducing domestic energy consumption while maintaining comfort.

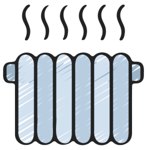
Students explore the levers of sobriety in housing through the example of a 100m² detached house built in the 1980s.

Reference situation - Maison Martin

- Heating: 15,000 kWh/year (gas)
- Hot water: 3,000 kWh/year (electric)
- Household appliances: 3,500 kWh/year
- Total: 21,500 kWh/year

Students calculate potential savings and assess the impact on daily comfort. They create a "home energy efficiency guide" with practical, illustrated tips.

Levers of sobriety by position



Heating (70% of consumption):

- Temperature reduction: 19°C instead of 21°C → -15%
- Programming/zoning: heat only occupied rooms → -20%
- Behavioral insulation: close shutters, avoid overheating → -10%

Hot water (15% of consumption):

- Reduced shower duration: 5 min instead of 8 min → -35%
- Water heater temperature: 55°C instead of 60°C → -8%
- Water saving installation → -20%

Household appliances (15% of consumption):

- Unplugging devices on standby → -10%
- Optimal use (full washing machine, short programs) → -20%
- Natural drying of laundry → -30% of the drying station

3. Food workshop: rethinking our food consumption

Objective: To explore the possibilities of reducing energy consumption linked to food.

Students could analyze the "food energy budget" of a typical family, with:

Reference situation - Typical food consumption

- Food transport: 2,000 kWh/year
- Refrigeration/Freezing: 1,500 kWh/year
- Cooking: 1,000 kWh/year
- Food waste: equivalent to 800 kWh/year

Levers of sobriety

Strategy	Description	Potential savings	Conditions for success
Circuits courts	Local and seasonal purchases	-40% transport	Local offer available
Anti-waste	Better inventory management	-50% waste	Meal planning
Economical cooking	Cooking optimization	-30% cooking energy	Change of habits
Sustainable food	More plant-based, less processed	-25% global	Cooking lessons

Conclusion & Further Reflexion

Exploring the levers of sobriety allowed students to discover that reducing energy consumption does not require sacrificing comfort, but rather intelligently optimizing uses. They understood that the possibilities for action vary greatly depending on territorial and socio-economic contexts, revealing inequalities in the face of energy sobriety. The analysis of the three areas (transport, housing, food) showed that the solutions systematically combine technical, behavioral, and organizational levers, with significant savings potential, but conditioned by social acceptability and available infrastructure.

The thematic workshop approach encourages students to take ownership of the issues and allows them to quantitatively measure the impact of their choices. The creation of "typical days" and practical guides develops their ability to project changes into everyday life. A comparative analysis of territorial constraints helps students move beyond a uniform vision of sobriety to understand the diversity of situations and possible solutions.

Students now understand the distinction between energy efficiency and sobriety, and are able to identify levers for action in their own life context. They have developed their understanding of the interdependencies between individual choices and structural constraints, as well as their ability to critically evaluate sobriety measures according to feasibility and acceptability criteria.

The students collectively identify the general principles of energy sobriety which emerge from these three workshops:

- Questioning automatisms: Do I really need this journey/this temperature/this purchase?
- Optimizing what already exists: How can I make better use of what I already have?
- Smart Planning: How can I organize my activities to be more efficient?
- Investing in efficiency: What equipment can reduce consumption?
- Collective solutions: How can certain needs be shared (transport, heating, household appliances, food purchases)?

To deepen the reflection and develop a more systemic understanding of the issues of sobriety, here are some key questions to explore collectively:

- **How can we overcome structural obstacles to energy efficiency?** An analysis of the infrastructure and public policies needed to democratize access to energy-efficient solutions. A reflection on the role of local authorities and businesses. An exploration of solidarity mechanisms to reduce inequalities in the face of the energy transition.
- **What are the risks and limitations of sobriety strategies?** Identifying potential rebound effects and social acceptability thresholds. Analyzing the tensions between individual sobriety and collective needs. Questioning the sustainability of behavioral changes over time.
- **How can we reconcile local sobriety with global challenges?** A reflection on the impact of individual choices on a European and global scale. Exploring the mechanisms for disseminating sober practices. Analyzing the coherence between local actions and global climate objectives.
- **What personal vision of energy efficiency should we build?** Preparation for critical self-evaluation of one's own practices. Reflection on the desirable balance between accepted constraints and preserved quality of life. Introduction to the prospective analysis of technological innovations that could facilitate or transform energy efficiency.

This synthesis prepares the personal and prospective reflection of the next phase.

Step 4 - Critical and prospective analysis of energy uses

Context and description of the problem to be solved at this stage: This final phase invites students to conduct an in-depth critical analysis of their own energy practices and to reflect on technological innovations in a forward-looking manner. Beyond a simple categorization of uses, students are encouraged to understand the sociocultural factors that influence their energy choices and to evaluate the transformative potential of emerging innovations from an energy transition perspective.

Learning Objectives: Critically analyze one's own energy uses, taking into account social and cultural contexts. Evaluate the potential and limitations of energy innovations from a systemic perspective. Construct a well-reasoned personal vision of energy efficiency and technological transition.

Conceptualisation

Research question: To what extent are our individual energy choices determined by sociocultural and structural factors rather than by rational decisions?

- Hypothesis: Individual energy uses result more from socio-economic constraints, cultural norms and available infrastructure than from deliberate choices based on a cost-benefit analysis, creating variable margins of action depending on the context.
- Key concepts:
  - Energy determinism: Influence of social, economic and technical structures on individual choices (reveals the limits of individual action)
  - Energy capital: Set of resources (economic, cultural, technical) enabling access to sustainable energy choices (explains inequalities in the face of sobriety)
  - Energy habits: Integrated and unconscious energy practices, transmitted socially (questions the automatic dimension of consumption)

Research question: How can emerging technological innovations structurally transform our lifestyles without creating new rebound effects?

- Hypothesis: Energy innovations have real transformative potential but their impact depends on their ability to modify existing socio-technical systems rather than simply improving the efficiency of current practices, which requires a systemic approach to their deployment.
- Key concepts:
  - Disruptive vs. Incremental Innovation: Distinction between technologies that transform systems vs. those that optimize them (determines transformation potential)
  - Socio-technical transition: Process of change simultaneously involving technologies, social practices and institutions (explains the conditions for the success of innovations)
  - Technology lock-in: Dependence on existing systems that hinders the adoption of alternatives (reveals barriers to transitions)

Research question: What articulation between individual action and collective transformation allows for an effective and socially acceptable energy transition?

- Hypothesis: The energy transition requires coherence between individual commitments and structural transformations: neither individual action alone nor imposed technical changes are sufficient, but their combination in collective dynamics can create virtuous circles of change.
- Key concepts:

- Multi-level coherence: Alignment between individual actions, technical innovations and public policies (conditions the effectiveness of transitions)
- Energy commitment: Form of citizen action combining personal practices and collective participation (goes beyond simple responsible consumption)
- Energy democracy: Citizen participation in technological choices and energy directions (questions the modes of governance of the transition)

Students Investigation

1. Personal energy audit and sociological analysis

Objective: Carry out a complete diagnosis of your own energy uses and analyze the factors that determine them.

Collection of personal data: Each student keeps a detailed energy diary (printable example available in the appendix) for one week, noting:

Day/Time	Energy use	Duration	Context	Possible choice?	Motivation for choice
Monday 7h	Hot shower	8 min	Press	Variable duration	Habit + comfort
Monday 7:15 a.m.	hair dryer	5 min	Alone	Yes/No	Social appearance
Monday 8 a.m.	Transport bus	25 min	With friends	Walking/cycling possible	Parents + weather
...	...	...	...	...	...

Advanced Quantification and Categorization: Students calculate their weekly consumption by item and project it over the year. They then use a more nuanced analysis grid:

Usage	Estimated consumption (kWh/year)	Status*	Room for maneuver	Constraining factors
Home-school transport	180	⚠ Constrained	Weak	Distance, limited public transport, parental permission
Room heating	800	⚠ Partially constrained	Average	Home insulation, family budget, minimum comfort
Digital entertainment	320	✓ Personal choice	Strong	Habits, social pressure, available alternatives
Lighting homework	45	✓ Necessary optimizable	Average	Schedules imposed, equipment available

Statuses:

- ***Constrained: Use determined by external factors (infrastructure, income, housing)***
- ***Partially constrained: Limited but existing room for maneuver***
- ***Personal choice: Usage determined primarily by individual preferences***
- ***Necessary to optimize: Essential but improvable use***

Analysis of sociocultural determinants: Students analyze the factors that influence their energy choices:

Influencing factor	Impact on my uses	Concrete example	Personal scope for action
Family income	Determines equipment and insulation	No air conditioning → fan	Zero in the short term
Social norms of my group	Influence of shower duration, clothing	"Normal" daily showers	Average (negotiation possible)
Territorial Infrastructure	Constrained modes of transport	No safe cycle path	Null individually
Housing facilities	Determines energy efficiency	Old electric radiators	Low (parental choice)
Energy education received	Influence level of consciousness	Knowledge of appliance consumption	Strong (continuous learning)

2. Prospective assessment of energy innovations

Objective: To analyze the transformative potential of energy innovations using a systemic approach.

Students work in teams to analyze an energy innovation in depth using a multidimensional evaluation grid (available for printing in the appendix):

A. Technical and economic potential

Criteria	Assessment	Justification
Technological maturity (scale 1-10)	___/10	Current state of development
Energy reduction potential	___%	Savings compared to current technologies
Deployment cost (€/inhabitant)	___€	Investments needed for generalization
Probable time frame for generalization	___years	Time for mass adoption

B. Impact on lifestyles

Aspect	Expected transformation	Probable acceptability
Habitat	Example: Energy-independent houses	✓ Strong / ⚠ Medium / ✗ Weak
Transport	Example: End of car ownership	✓ Strong / ⚠ Medium / ✗ Weak
Work	Example: New technical professions	✓ Strong / ⚠ Medium / ✗ Weak
Hobbies	Example: Multiple urban green spaces	✓ Strong / ⚠ Medium / ✗ Weak

C. Systemic effects and risks

Dimension	Opportunities	Risks
Environmental	Ex: Reduction of carbon footprint	Ex: New needs in raw materials
Social	Ex: Democratization of access to energy	Ex: Digital exclusion, jobs cut
Economic	Ex: New sectors of activity	Ex: Transition costs, obsolescence
Geopolitics	Ex: Energy independence	Ex: New technological dependencies

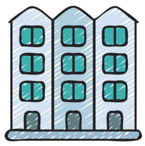
Each team has 2 hours to:

1. **Search for reliable data on their innovation (institutional sites, scientific reports, prospective studies)**
2. **Complete the evaluation grid by justifying each evaluation**
3. **Identify 3 critical conditions for successful innovation**
4. **Formulate an adoption scenario (optimistic, pessimistic, realistic)**

Innovations to analyze

Team 1: Shared Autonomous Electric Vehicles

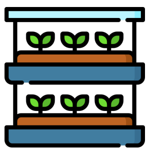
Team 2: Connected positive energy buildings (BEPOS+)



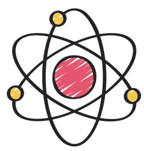
Team 3: Smart Grids with Distributed Storage

Team 4: Green hydrogen for industry and heavy transport

Team 5: Automated Urban Vertical Agriculture



Team 6: Commercial Nuclear Fusion



3. Prospective debate and construction of personal visions

Objective: Compare analyses and construct a personal, reasoned vision of the desirable energy transition.

Each team presents its innovation in 3 minutes according to the format:

- **Technical presentation: What is innovation?**
- **Transformative Potential: What Changes in Our Lives?**
- **Conditions for success: What obstacles must be overcome?**

Students circulate between "booths" where each team presents their work interactively. They record their questions and evaluations of each other's innovations.

Cross-assessment grid

Innovation	Excites me because...	Worries me because...	Probability of success (/10)
Autonomous vehicles	_____________________	_____________________	____/10
BEPOS+ buildings	_____________________	_____________________	____/10
[...]	_____________________	_____________________	____/10

It is possible to organize a debate to close the last phase around the following questions:

- **Question 1: "Should we prioritize innovations that radically transform our lifestyles or those that preserve them?"**
    - “Transformation” arguments: maximum efficiency, adaptation to challenges
    - "Preservation" arguments: social acceptability, cultural continuity
- **Question 2: “How can we reconcile technological innovation and energy efficiency?”**
    - Risk of rebound effect (more efficiency → more use)
    - Possible complementarity (technologies + behavioral changes)
- **Question 3: “Who should decide on technological directions?”**
    - Democratic legitimacy vs. technical competence
    - Role of lobbies and economic interests

Conclusion & Further Reflexion

This final phase allowed students to understand that their individual energy choices are part of a complex network of sociocultural and structural determinants that limit their decision-making autonomy. The personal audit revealed to them the extent of the constraints weighing on their energy practices, going far beyond simple individual preferences. The prospective analysis of innovations gave them the tools to critically evaluate technological promises, distinguishing between disruptive innovations and incremental improvements and identifying the systemic conditions necessary for their success.

The personal audit approach promotes a reflective and guilt-free awareness of energy practices. The prospective analysis of innovations develops critical thinking in the face of technological discourse and strengthens the ability to argue on complex subjects. The structured debate allows students to compare their visions and develop nuanced positions on the challenges of the energy transition. The final personal synthesis consolidates learning by anchoring knowledge in a coherent life plan.

Students have developed a systemic understanding of energy issues that goes beyond the simplistic opposition between individual responsibility and structural constraints. They have mastered a method for prospectively evaluating technological innovations and are able to formulate a well-reasoned personal vision of their role in the energy transition. They have acquired the conceptual foundations necessary to participate in an informed manner in energy debates as future citizens.

To deepen the reflection and open up perspectives for citizen action, here are some key questions to explore collectively:

- **How can we transform identified structural constraints into levers for collective action?** Exploring mechanisms for citizen mobilization around energy issues. Analyzing the role of associations, communities, and businesses in creating alternatives. Reflecting on the conditions for the emergence of territorial energy transition dynamics.
- **What democratic tools allow citizens to influence technological and energy choices?** An exploration of citizen participation bodies (development councils, consensus conferences, participatory budgets). An analysis of energy democracy experiences in Europe. A reflection on the civic skills needed to participate in technical debates.
- **How can we reconcile the climate emergency with the social acceptability of transitions?** Exploring the differentiated pace of change across social groups and territories. Analyzing social justice mechanisms in the

energy transition. Reflecting on the balance between incentives, constraints, and support in public policies.

- **What personal contribution can you make to the energy transition beyond individual actions?** Identifying forms of civic engagement accessible to young people (associations, school projects, local initiatives). Exploring the links between training/career choices and energy issues. Building a long-term vision of your role in tomorrow's energy society.

This phase concludes the protocol by giving students the conceptual and methodological tools to become informed actors in the energy transition, capable of articulating critical analysis, prospective vision and personal commitment in a coherent civic approach.

Multidimensional assessment grid

Student Name:	Date:

A. Technical and economic potential

Criteria	Assessment	Justification
Technological maturity (scale 1-10)	___/10	
Energy reduction potential	___%	
Deployment cost (€/inhabitant)	___€	
Probable time frame for generalization	___years	

B. Impact on lifestyles

Aspect	Expected transformation	Probable acceptability
Habitat		☐ Strong ☐ Medium ☐ Weak
Transport		☐ Strong ☐ Medium ☐ Weak
Work		☐ Strong ☐ Medium ☐ Weak
Hobbies		☐ Strong ☐ Medium ☐ Weak

C. Systemic effects and risks

Dimension	Opportunities	Risks
Environmental		
Social		
Economic		
Geopolitics		

Logbook

Student Name:

Date: from ..../..../........ to ..../..../........

Day/Time	Energy use	Duration	Context	Possible choice?	Motivation for choice

Print as many sheets as needed to collect information for 1 week

Analysis of uses

Student Name:

Date: from ..../..../........ to ..../..../........

Usage	Estimated consumption (kWh/year)	Status*	Room for maneuver	Constraining factors

**Statuses:*

- ***Constrained: Use determined by external factors (infrastructure, income, housing)***
- ***Partially constrained: Limited but existing room for maneuver***
- ***Personal choice: Usage determined primarily by individual preferences***
- ***Necessary to optimize: Essential but improvable use***

Print as many sheets as necessary to analyze all uses